NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME: **Mark Scheme**

CLASS:

BIOLOGY

9648/03

Paper 3 Applications and Planning

September 2015

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	60
For Examiner's Use	
4	12

This document consists of **15** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 **Fig. 1.1** outlines how a gene coding for human insulin is produced by genetic engineering techniques.

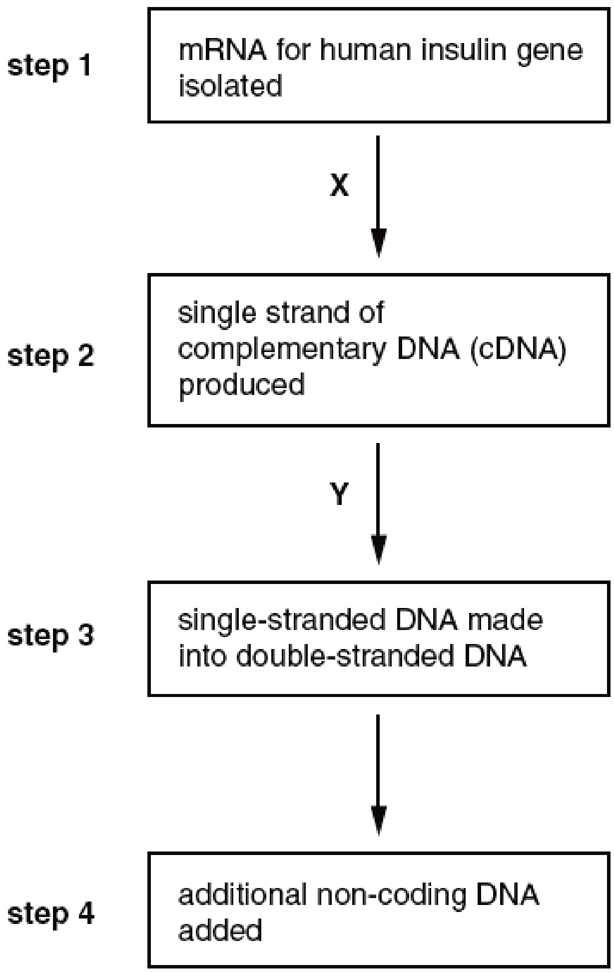


Fig. 1.1

(a) **(i)** Name the enzymes X and Y.

X reverse transcriptase

Y DNA polymerase

[2]

(ii) Explain why the starting point in this procedure is mRNA.

Large number of copies of mRNA readily available ;

idea of mRNA is only from gene coding for insulin (being expressed) ;

easier than, extracting/locating, gene from cell's DNA ;

AVP ; e.g. introns already removed/bacteria cannot remove introns ;

[2]

(iii) Suggest why some methods of manufacturing genetically engineered insulin use eukaryotic yeast cells rather than prokaryotic bacterial cells.

in yeast cells promoters already present ;

have rER/Golgi body ;

so, insulin can be modified/insulin is in correct 3D conformation ;

AVP ; e.g. ref. to YAC holding more DNA than BAC ;

[2]

(b) The artificial plasmid, pBR322, was constructed to act as a vector. It has often been used to insert human genes, such as the human insulin gene, into the bacterium, *Escherichia coli*.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, *Bam*HI, in the middle of the tetracycline resistance gene.

A pBR322 plasmid was cut using *Bam*HI and the cDNA gene for human insulin inserted into it.

Fig. 1.2 shows pBR322 and the recombinant plasmid.

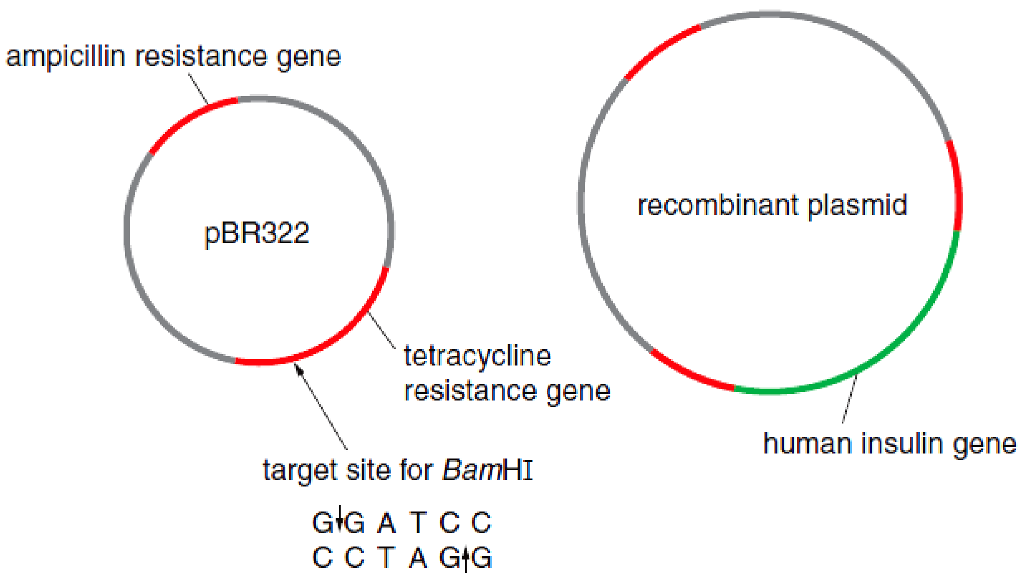


Fig. 1.2

With reference to **Fig. 2.1**, describe how a cDNA human insulin gene can be inserted into pBR322 that has been cut by *Bam*HI.

ref. sticky ends GATC and CTAG ;

complementary bases (pairing) between A to T and C to G ;

Hydrogen bonds formed to sticky ends of plasmid ;

formation of phosphodiester bonds to seal the gaps in sugar-phosphate backbones by DNA ligase ;

AVP ; ref. terminal transferase

[3]

(c) Bacteria were then mixed with the recombinant plasmids. Those bacteria which had successfully taken up recombinant plasmids were identified using the following steps:

step 1 – the bacteria were spread onto culture plates containing nutrient agar and ampicillin and incubated to allow colonies to form

step 2 – some bacteria from each of the colonies growing on these plates were transferred to plates containing nutrient agar and tetracycline, as shown in **Fig. 1.2**.

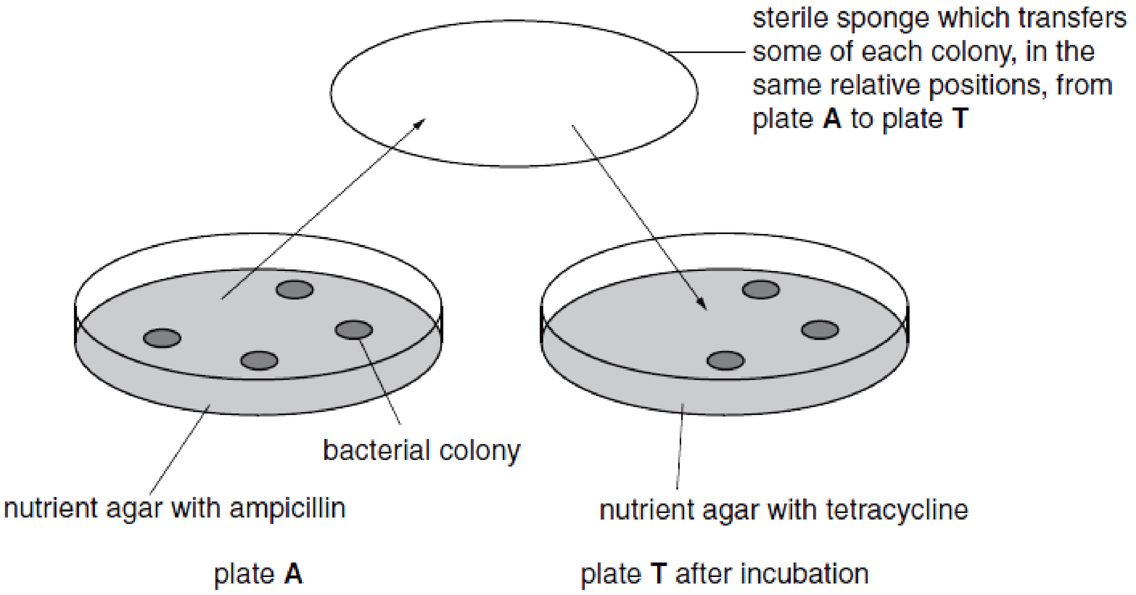


Fig. 1.3

(i) Explain why the bacteria were first spread onto plates containing ampicillin.

idea of identifying bacteria that, are transformed / have taken up plasmid / have taken up ampicillin resistance gene;

these bacteria have survived;

these bacteria may contain pBR322 or recombinant plasmid / plasmids taken up may not contain human insulin gene;

other untransformed bacteria have been killed;

[2]

(ii) Explain why it is important, for identifying bacteria that have successfully taken up the recombinant plasmid, that on pBR322 the target site for *Bam*HI is in the middle of the tetracycline resistance gene.

(BamHI) breaks the tetracycline resistance gene / insertional inactivation;

inserting human insulin gene makes tetracycline resistance gene inactive ;

colonies that are ampicillin-resistant but not tetracycline-resistant have taken up recombinant plasmid / insulin gene ;

colonies that survive on, tetracycline / both ampicillin and tetracycline / plate T, have not taken up the recombinant plasmid / insulin gene ;

[3]

(iii) Use a label line and the letter **C** to identify, on **Fig. 1.3**, a colony of bacteria that contain the recombinant plasmid.

Answer on Fig. 1.2 left hand colony on plate A ;

[1]

[Total: 15]

2 **Fig. 2.1** shows the inheritance of SCID in a family in which both parents are carriers. Only one of the sons suffers from SCID.

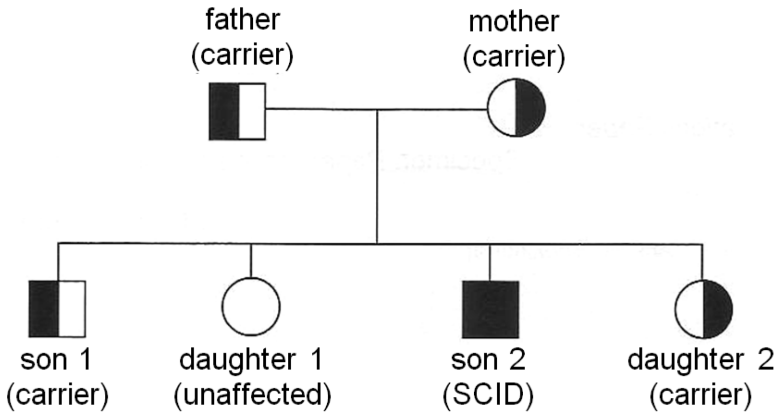


Fig. 2.1

Two RFLP morphs have been found tightly linked to the adenosine deaminase (ADA) gene which can be used for genetic screening. A radioactive probe has been designed to reveal the two RFLP morphs.

Both RFLP morphs and radioactive probe are shown in **Fig. 2.2** with the arrows representing the *Pvu*II restriction sites. The asterisk (*) indicates the position of a mutation at the restriction site that results in the loss of the restriction site. Morph 2 was found to be linked to the mutant ADA gene.

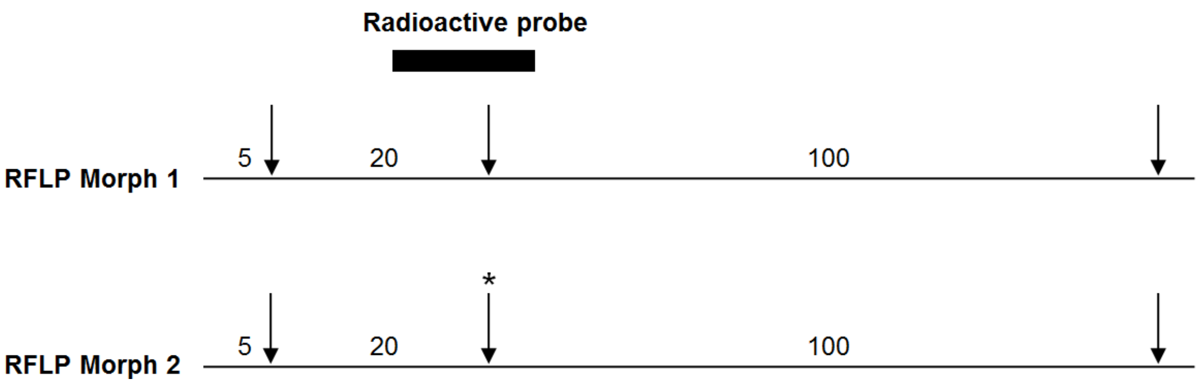
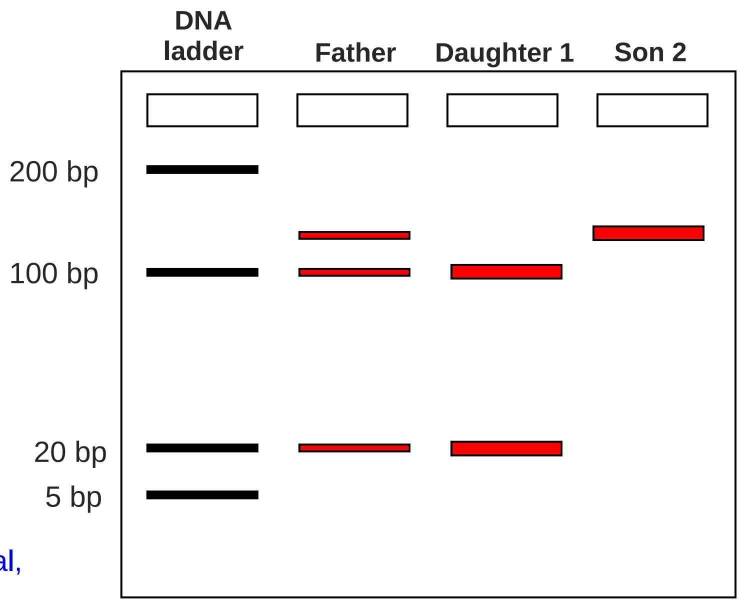


Fig. 2.2

(a) **(i)** With reference to **Fig 2.1** and **Fig. 2.2**, draw the resulting autoradiogram you would expect to see in the space below.



For each individual,
1 Correct bands
2 Correct thickness

[3]

(ii) Explain the role of the DNA ladder.

1 as a reference/standard / providing known DNA fragments of specific lengths

2 to estimate the size of unknown DNA molecules

[1]

(b) Gene therapy can be used to treat severe combined immunodeficiency disease (SCID). A deficiency of adenosine deaminase (ADA) is a cause of severe combined immunodeficiency syndrome (SCID). Affected individuals are unable to produce ADA. Without this enzyme, T lymphocytes (a type of white blood cell) cannot provide immunity to infection. **Fig. 2.3** shows the processes involved in the treatment of ADA deficiency by gene therapy.

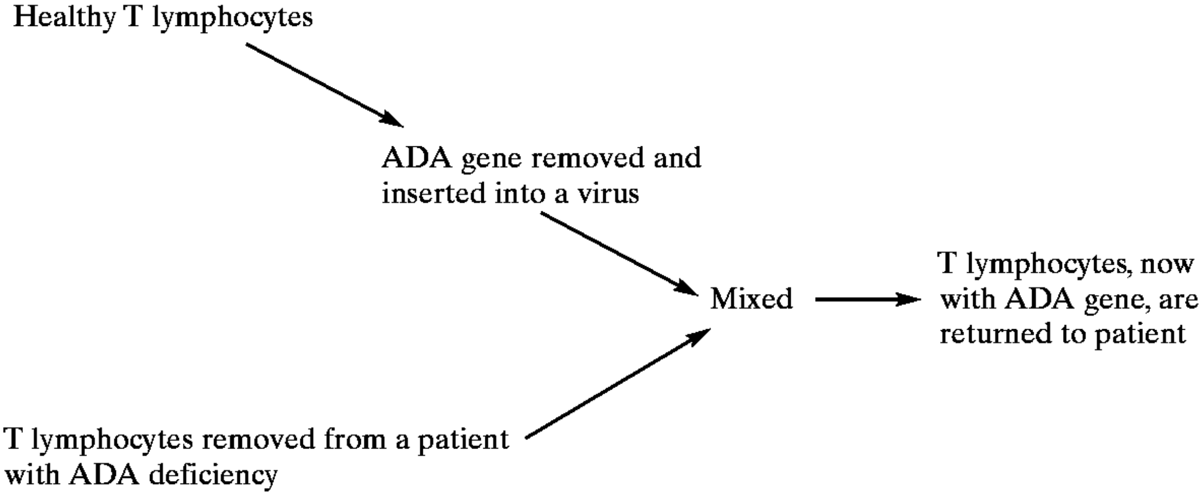


Fig. 2.3

(i) Explain why individuals who have been treated by this method of gene therapy do not pass on the functional ADA allele to their children.

Reproductive <u>gametes</u> of these individuals who have been treated will not contain the ADA allele/gene;

Treatment targets at the <u>somatic cells</u> and <u>not germ cells</u> / <u>gametes</u> so it will not get inherited;

[2]

(ii) Suggest the advantages of a viral gene delivery system.

able to deliver genes to specific target tissue;

high transfection efficiency;

able to integrate genes into genome / delivers gene into nucleus, allows stable propagation / reduce chance of cytoplasmic degradation;

[2]

(iii) Suggest why the ex vivo method was used in this case.

T-lymphocytes can be easily transfected and cultured in the laboratory;

Also allow for selection of successfully transformed cells;

Prevent the retrovirus from targeting other rapidly proliferating cells;

Reject: viral vector can regain ability to cause disease

[1]

(iv) Other than the short-lived nature of the treatment, explain other problems encountered in using gene therapy as a treatment for genetic diseases, such as SCID.

idea of inappropriate immune response to viral vectors ;

gene inserted into the wrong place inducing a tumour ;

another problem associated with gene being inserted in, wrong place / into another gene ;

child receiving treatment for SCID developed leukaemia ;

further detail regarding treatment for SCID ;

credit a case study ;

AVP ;

[3]

(v) Name another method, other than gene therapy, which can be used for the treatment of ADA-deficiency.

bone marrow transplantation for a perfectly matched sibling donor;

enzyme replacement therapy / weekly intramuscular injections of ADA;

[1]

[Total: 13]

3 Plant tissue culture has many commercial applications. These include the production of large numbers of genetically identical plants and the culturing of plant cells in a bio-reactor for the production of secondary products.

Early attempts to culture plant tissue on a culture medium were not always successful.

The culture medium contained a range of substances that were known to be requirements for plant cell growth and division. These included:

- elements, present as mineral ions
- a sugar suitable for plant cells to use

(a) Explain why a sugar was included in the plant tissue culture medium.

Photosynthesis not yet occurring or at low rate, therefore sugars not being synthesized;

Sugars required as energy source for respiration/ATP production/respiratory substrate OR

Source of carbon for carbohydrates like starch, sucrose, and organic molecules proteins or enzymes/other suitable named organic molecules;

Max 2 marks

[2]

Investigations have shown that the original culture medium lacked key components called plant growth regulators needed for successful plant cell growth and division.

Although early attempts to culture plant tissue were not always successful, some investigators have reported successful growth using root tips.

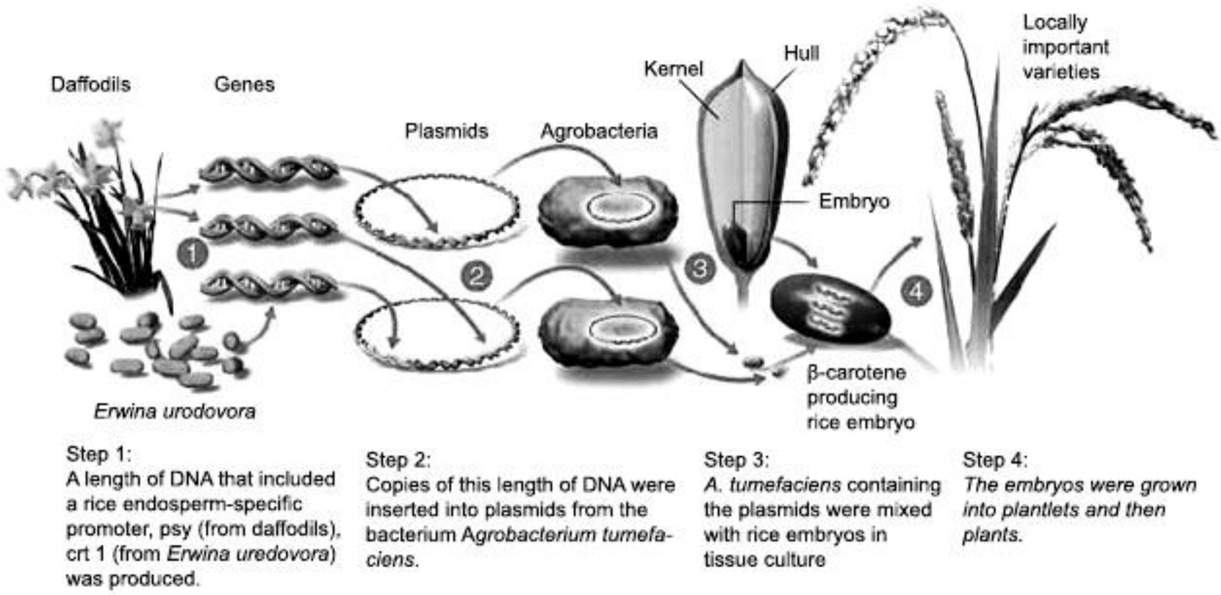
(b) Suggest one reason why plant tissue culture was more successful using organs such as root tips.

Meristematic/mitotic tissue; reference to cell division activity in root tips;

Root tip region still immature tissue; Still producing plant growth regulators e.g. auxins, cytokinins etc;

[1]

(c) Rice plants produce the precursor of vitamin A, β-carotene, in their green tissues, but not in the edible endosperms of their seeds because the seeds lack the enzymes for two steps of the pathway for β-carotene production. Genetic engineering methods and plant tissue culture techniques have been used to insert the genes for these enzymes into the rice embryos to produce rice seeds that contain β-carotene. These rice seeds are known as Golden Rice™. The production of such rice is an important step towards alleviating vitamin A deficiency in many rice-based societies in South-east Asia.



(i) Explain why *Agrobacterium tumefaciens* containing plasmids were used in Step 3.

Vector that can insert DNA into plant DNA/ carries genes into plant genome;

Contains Ti-DNA that allows insertion of DNA into host genome;

[2]

(ii) Besides Agrobacterium-mediated gene transfer, state one method that is commonly used to insert DNA into cells of monocotyledonous plants.

Microprojectile/ gene gun/ micropipette/ biolistics;

[1]

(d) The concentration of β-carotene in the seeds from the field trial of the original strain of Golden Rice™ in 2004 was only about 6μgg^{-1}. The rate limiting step in the production of β-carotene was found to be the activity of the enzyme encoded by the *psy* gene. Suggest what is meant by a *rate limiting step* in a metabolic pathway.

Step with the lowest rate;

Slows the entire reaction/ determines the rate of rest of the pathway/ step that controls other steps;

Step that controls rate at which product is made;

[2]

(e) The concentration of β-carotene was increased by breeding with other varieties of rice and by further genetic engineering. More recent strains, such as Golden Rice 2™, contain about 31μgg^{-1} of β-carotene in the endosperm of the seeds.
Suggest what further genetic engineering was carried out to increase the concentration of β-carotene in the seeds.

Increase expression of psy ;

By genes from other sources;

By using a more efficient/ better promoter;

Inserting more than one copy of gene;

[2]

(f) Field trials of genetically engineered crops, such as Golden Rice™, hope to identify risks to the environment or to human health of growing and eating the crops.

Suggest

(i) one possible risk to the environment of growing a genetically engineered crop.

one possible risk to the environment of growing a genetically engineered crop;

contamination of other crop, with reference to organic crops;

gene transfer to wild relative;

gene transfer via bacteria or viruses with unknown effect;

recipient plant outcompetes others/ ref to superweed;

toxic to other pests;

[1]

(ii) one possible risk to human health of eating a genetically engineered crop.

allergy;

long term toxicity;

antibiotic resistance of gut bacteria if antibiotic marker is used;

gene transfer to, virus to human cells/ gut bacteria, unknown effect (if idea is not already given in part (i))

[1]

[Total: 12]

Planning Question

4 You are required to plan an investigation to determine how the age of the mealworms affect their rates of respiration.

A bicarbonate indicator is a type of pH indicator that is sensitive enough to show a colour change as the concentration of carbon dioxide gas in an aqueous solution changes.

increasing CO_2 in indicator ←			atmospheric CO_2 level			decreasing CO_2 in indicator →		
yellow		orange		red		magenta		purple
pH 7.6	pH 7.8	pH 8.0	pH 8.2	pH 8.4	pH 8.6	pH 8.8	pH 9.0	pH 9.2

Bicarbonate indicator is red in equilibrium with atmospheric air. It changes from red to magenta to deep purple as carbon dioxide concentration decreases.
It changes from red to orange to yellow as carbon dioxide concentration increases

You are provided with the following equipment which you may use or not in your plan, as you wish.

- a supply of different day old mealworms
- syringes
- delivery tubes
- rubber tubing
- forceps
- bicarbonate indicator solution
- potassium hydroxide (to absorb atmospheric carbon dioxide)
- Vaseline petroleum jelly
- stopwatch
- colorimeter and cuvettes
- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.

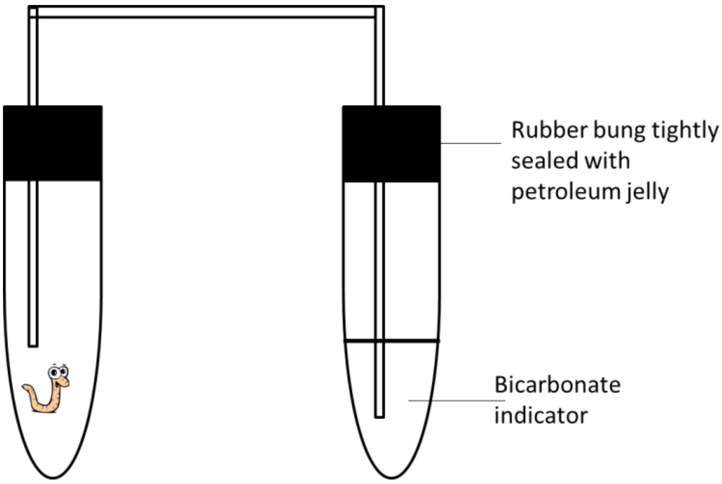
Your plan should have a clear and helpful structure to include
- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method, a control experiment and any recommended safety measures
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagram/s
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

Suggested Answer:

- Respiration and rate of carbon dioxide release
  - Reference to decarboxylation during Krebs cycle, release carbon dioxide
- Factors affecting rate of respiration
  - rate can be affected by age of the organism. A older mealworm may possess more cells and thus more energy required to sustain respiratory activity.
- Measurement of carbon dioxide released using colour as an indicator
- Independent variable is the age of mealworm used. The dependent variable is the distance moved by the manometer fluid.
- Hypothesis: The larger the mass, the lower the pH/ more yellow the colour

Recommended Procedure



1. Place a 1 day-old mealworm into a test tube and replace the rubber bung with the delivery tube.
2. Place the other end of the delivery tube into another test tube filled with $5cm^3$ of bicarbonate indicator.
3. Ensure that both test tubes are tightly sealed using petroleum jelly around the rubber bungs.
4. Allow the set up to adjust for 3 minutes before starting experiment.
5. Calibrate the colorimeter to green light using the solution from the control test tube.
6. Transfer 1ml of the bicarbonate solution into a cuvette after 20min and measure the colour intensity of the solution.
7. Repeat steps 1-6 for mealworms of 4 other ages.
8. Repeat steps 1-7 two more times to obtain a total of 3 readings to calculate the mean colour intensity.
9. Set up a control test tube using steps 1-7, replacing the mealworm with a glass bead of similar mass.
10. Record readings in a table.
11. Plot a graph of percentage colour intensity against age of mealworms.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Compare replication in cells and the polymerase chain reaction (PCR). [8]

(b) Outline the basis and process of gel electrophoresis. [7]

(c) Describe the features of blood stem cells and explain their normal functions. [5]

[Total: 20]

Similarities

1. Involve separation of two polynucleotides strands and each used as template;
2. Require DNA polymerase and primers;
3. Involve synthesis of new strands in 5" to 3" direction **or** primers added at the 3" ends **or** nucleotides are added to the 3" OH ends of the primers;
4. Both processes resulted in an increase in the amount of DNA

Features	**DNA replication**	**PCR**
Separation	Separation of double helix by **helicase**	Separation by heating to **high temperature / denaturation**;
Replicated section	**Entire** DNA molecule replicated	A **section** of DNA replicated;
Primers used	**RNA** primers	**DNA** primers;
Presence of lagging strands	Presence of Okazaki fragments **or** supercoiling requiring the use of topoisomerase	No Okazaki fragments **or** lack of supercoiling not requiring the use of topoisomerase;
Polymerases involved	**DNA** polymerase	**Taq** DNA polymerase needed;
Accuracy of replication	**Proofreading** ability of DNA pol	Taq polymerase **lacks proofreading** ability;
Start of replication	**Origins of replication**	Where **primers binds**;
Synthesis of Primers	Primers are made by **primase or** primers excised	Primers are added / made in the lab **or** primers not excised;
Temperature	Occurs at body temperature	Annealing 50-60°C Elongation 72°C and denaturation 90°C
Aim / Purpose	Occurs prior to cell division to replace dead cells lost due to wear and tear	To amplify DNA for forensics analysis

Agarose gel electrophoresis is a method for separating and visualizing DNA fragments produced by restriction digestion of DNA.
The fragments are separated by charge and size by forcing them to move through a agarose gel matrix which is subjected to an electric field.

Principles of agarose gel electrophoresis

- Biological molecules such as DNA, RNA and proteins are electrically charged particles at a given pH.
  - DNA is negatively charged due to the phosphate groups of the sugar-phosphate backbone.
  - If the sample contains DNA fragments of different sizes, they can be separated based on their sizes when they are subjected to an electric field.
  - Smaller fragments will move faster then the larger fragments.

Step 1: Casting of the agarose gel

- Agarose is a polysaccharide derived from seaweed that forms a gel when dissolved in an aqueous solution.
- Agarose solution will be poured into casting tray and allowed to solidify.
- With the help of the comb, small indentations called wells are formed at one end of the gel, where DNA samples can be loaded.
- Each well corresponds to one lane.

Step 2: Loading of sample

- DNA samples are first mixed with a loading dye which increases its density and helps the DNA to sink to the bottom of the well.
- Loading dye also helps to monitor the progress of the electrophoresis as DNA is invisible.
- Markers are usually run in the 1st lane. Markers are mixtures of DNA fragments of known sizes. By comparing the markers' fragment against the unknown fragments, we can estimate the sizes of the unknown fragments.
- Samples are loaded into wells with the help of a micropipette.

Step 3: Running of gel

- Current is switched on.
- Negatively charged DNA fragments will migrate out of the well and are attracted to the positive electrode.
- Separation of the fragments will be based on their sizes and can be seen as discrete bands on the gel.

Step 4: Visualization

- Before the band reaches the end of the gel, current will be switched off.
- Gel is then stained with DNA binding dye (ethidium bromide which is carcinogenic).
- Separated DNA fragments can been clearly under the UV light.

1. type of adult stem cell;
2. found in bone marrow of long bones;
3. capable of dividing and renewing themselves for long periods;
4. unspecialized, does not have tissue specific structures for them to perform specialized functions;
5. can give rise to specialized cell types;
6. multipotent, can differentiate into a limited range of cell type, usually of a closely related family of cells;
7. e.g .red blood cells, white blood cells, platelets;
8. primary function is to maintain the steady state of functioning of a cell by generating replacement for cells lost through disease, tissue injury or normal wear-and-tear;